TECHNICAL WHITEPAPER

HYPERLIQUIDFLOW

Rules-based smart money intelligence for Hyperliquid perpetuals.
No ML. No paid data. Public API, end to end.

VERSION	PUBLISHED	EXCHANGE
2.0	April 2026	Hyperliquid Perps

DOCUMENT

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SECTION 1

Problem Statement

Why 1.6 million wallets is a data problem, not a data opportunity

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Hyperliquid publishes the complete position and fill history of every wallet on its platform through a free public API. It is one of the most transparent on-chain trading environments in existence. It is also, without a filter, completely unreadable at scale.

TOTAL WALLETS	DAILY CANDIDATES	ACTIVE COHORT
~1.6M	~4,500	~500
On Hyperliquid	Processed per scan	Passing all quality gates

The noise problem

The public dataset contains every category of participant: losing retail accounts, bot farms, exchange hot wallets, wash traders, and dormant addresses, alongside a small population of genuinely skilled, risk-adjusted traders. There is no built-in filter. Reading raw position data across 1.6 million wallets without one is reading noise.

The traders who matter are in that dataset. They are simply outnumbered by a factor of several thousand.

The signal problem

Even if you identified the right wallets manually, tracking their position changes in real time is a continuous engineering problem. A single wallet move means nothing in isolation. The valuable information surfaces when several proven wallets shift position in the same direction on the same asset within a short window, or when a high-conviction wallet enters a trade aligned with a funding-rate carry opportunity. Detecting those patterns requires comparing consecutive snapshots across a verified population, not one-time lookups.

What HyperliquidFLOW solves

HyperliquidFLOW is the filter and the detector. It narrows 1.6 million wallets to a verified cohort of approximately 500, watches that cohort on a continuous cycle, and fires a

structured signal when their collective or individual behavior matches one of 13 rule-based patterns. Each signal is a factual statement about observed behavior, paired with a confidence score derived from the signaling wallet's own 30-day track record.

No predictions. No interpretations. A statement of what happened and a probability estimate of whether similar past behavior has been profitable.

SECTION 2

Approach

Three design constraints, applied to every part of the system

Rules, not models

Every decision in the system is a deterministic rule readable in the codebase. Wallet activation, deactivation, signal detection, and confidence scoring are all rule-based. There is no machine learning, no hidden embedding, no model to audit separately from the code.

A signal either fires or it does not. A wallet either passes the activation gates or it does not. The criteria are public, versioned, and stable between scans. This makes the system falsifiable: if a recipe consistently misfires, the rule can be tightened. If an activation gate cuts too many good wallets, the threshold can be adjusted. The outcome tracking pipeline measures results automatically, removing the need for manual audits.

Public data only

All inputs come from the Hyperliquid public API: fill history, current positions, funding rates, and order book data. No paid feeds. No proprietary order flow. No data vendors. Every observation in the system is independently verifiable by anyone with API access.

The only external enrichment is the Hypurrscan public label index, which surfaces advisory name tags for known addresses. These labels appear in the UI as context but do not affect scoring or activation.

Single-exchange focus

HyperliquidFLOW does not attempt to aggregate signals across multiple venues. Wallet activity on Hyperliquid is treated as the complete picture. This is both a constraint and an advantage: the dataset is consistent, latency is uniform, and the fill format is predictable. The manipulation detector mitigates the most common edge case: wallets with suspiciously offsetting internal positions are removed from the cohort before they can distort signals.

SECTION 3

Cohort Construction

How ~4,500 daily candidates become ~500 trusted wallets

The active cohort is the foundation of everything downstream. Every signal, score, and UI surface operates on it. The quality of the cohort determines the quality of the output.

3.1 Discovery

Each day at 00:00 UTC, the pipeline collects candidates from two independent sources:

- **Leaderboard.** The public Hyperliquid leaderboard, paginated to collect top-ranked accounts by 30-day PnL.
- **Fills mining.** Wallets appearing in recent fills for the top-10 coins by 24h volume. This catches active traders absent from the leaderboard because their total PnL is modest, but whose fill frequency and win rate qualify them.

A typical run yields approximately 4,500 candidates before any filtering begins.

3.2 Activation Gates

Each candidate is evaluated against nine independent quality gates. All nine must pass for activation. A single failure rejects the wallet, and the rejection reason is recorded so the funnel is auditable.

GATE	CRITERION	PURPOSE
Win rate	>52%	Removes coin-flip traders
Trade frequency	≥5 trades/week (30d avg)	Ensures sufficient sample size for scoring
Account equity	>$1K	Excludes dust accounts
Max drawdown	<60%	Removes blow-up-prone accounts
Risk-adjusted return	Positive (Sharpe proxy >0)	Requires positive risk-adjusted return
Regime alignment	Structural gate	Structural check, not a directional requirement
Trading recency	≥3 fills in last 7 days	Excludes dormant accounts
Manipulation flag	None	Removes coordinated self-trades
History depth	≥10 days with realized PnL	Requires statistical depth for scoring

APRIL 2026 SCAN RESULT

The 2026-04-18 run activated 493 wallets from 4,496 candidates, a 62% reduction from the prior baseline of approximately 1,285 wallets recorded before these gates were introduced. All nine rejection categories fired, confirming the gates are not concentrated in a single criterion.

THE 500 WALLETS

SIMPLE

Imagine 1.6 million people are fishing in the same lake. We found the 500 who catch fish consistently, week after week, and we watch what bait they use. When several of them switch to the same new lure at the same time, that is worth knowing about.

TECHNICAL

From ~4,500 daily candidates, wallets are activated by passing 9 gates: win rate >52%, ≥5 trades/week over 30 days, equity >$1K, max drawdown <60%, positive risk-adjusted return, recency of at least 3 fills in the last 7 days, at least 10 days with realized PnL, and no manipulation flag. April 2026: 493 activated from 4,496.

3.3 Continuous Hygiene

Activation happens once daily. Three continuous processes run between scans to keep the cohort clean:

Realtime health monitoring. Every live cycle, the pipeline checks whether an active wallet has gone quiet (no positions for 48 hours), blown up (equity below minimum floor), or

stopped trading (no fills in 7 days). Failures trigger immediate deactivation, not a deferred queue for the next scan.

Manipulation detection. Runs during the daily scan across the last 30 days of fills. Wallets trading with themselves, or clustering into rings where member A's buys correlate with member B's sells at unusual rates, are flagged and removed.

Behavior profiling. For each active wallet, the pipeline builds a profile: trading style (scalper, swing, position), conviction level (notional concentration in top position), and regime tendency (long-bias, short-bias, neutral). Profiles feed the ranking layer and UI detail pages but do not gate activation.

A wallet in the active cohort has survived all three continuously, not just at activation.

3.4 Composite Scoring

Activated wallets are ranked by a four-factor composite score, each component normalized to a 0-to-1 range:

- **Risk-adjusted return (35%).** Mean daily PnL divided by its standard deviation over 30 days, normalized so a raw Sharpe of 3.0 maps to 1.0.
- **PnL consistency (25%).** Fraction of 7-day windows in the last 30 days with a positive sum.
- **Drawdown score (25%).** One minus the worst cumulative drawdown from the 30-day PnL curve.
- **Regime fit (15%).** Alignment between the wallet's current directional bias and the BTC market regime.

All inputs come from 30 days of realized daily PnL, built from closing fills only. Unrealized PnL is never counted. Wallets scoring below 0.35 are pruned at the next cycle, except the top four by score, which are always retained to prevent simultaneous mass attrition.

> **KNOWN LIMITATION**
>
> The 35/25/25/15 weight split is an engineering assumption, not an empirically optimized result. A weight-validation script correlates scores against forward EV performance, but the weights have not been fit to that data. Calibration is on the roadmap.

SIMPLE

Each of our 500 wallets gets a report card. How consistent are their returns? How bad were their worst days? Are they trading in the right direction for the current market? The total is a number from 0 to 1. Higher is better.

TECHNICAL

Four-factor composite: risk-adjusted return (35%), PnL consistency (25%), drawdown score (25%), regime alignment (15%). All inputs from 30 days of realized daily PnL, closing fills only. Wallets below 0.35 pruned at next cycle; top-4 by score always retained.

SECTION 4

Signal Detection

13 rule-based behavioral recipes, applied every cycle

4.1 Recipe Framework

Each cycle, the pipeline takes two consecutive snapshots of cohort positions and passes the diff to 13 independent detection recipes. Each recipe receives a pair of position snapshots and returns zero or more signal events. The interface is uniform: a new recipe written to the same pattern can be registered without modifying the rest of the detection pipeline.

Recipes operate at two levels:

- **Per-wallet recipes** run on one wallet's snapshot pair at a time. Signals carry confidence scores derived from that wallet's 30-day backtest.
- **Cohort-level recipes** aggregate across all active wallets. They fire on group behavior: concentration, coordination, and funding divergence. Confidence scores are not applicable for these signals, since there is no single wallet track record to reference.

4.2 The 13 Recipes

Momentum

- **Whale Convergence.** ENTRY Three or more wallets add combined ≥$500K in the same direction within a 5-minute window.
- **Coordinated Exit.** EXIT Three or more wallets reduce the same coin by combined ≥$500K within 5 minutes.
- **Dip Conviction.** SCALE IN A wallet with score ≥0.65 adds after the coin drops 2× its 4-hour range from the 4-hour high, clamped to [6%, 15%].

Structural Risk

- **Silent Loading.** ALERT Two or more wallets (score ≥0.60) each add >$25K into a coin flat (<0.5%) for 30 minutes with a liquidity buffer below 10%. 20-minute cooldown per coin.

- **Crowded Coin.** ALERT One coin exceeds 60% of total cohort notional (70% threshold for ETH; BTC excluded).

Carry and Funding

- **Funded Edge.** ENTRY New position aligned with positive funding (>0.03%/hr), gated on at least 10 prior signals and a positive win rate for this recipe. Cold-start suppressed.
- **Smart Money vs. Retail.** ALERT Cohort direction opposes the funding rate sign when funding exceeds 0.05%/hr.
- **Funding Surge.** ALERT Funding rising for three or more consecutive cycles and crossing 0.03%/hr.

Liquidation and Cascade

- **Liquidation Flush.** ALERT Cohort notional on a coin drops >5% while price moves >1.5% (BTC/ETH) or >3.5% (alts) versus the prior cycle. Best-effort approximation. See Section 7.

Quality Filters and Lifecycle

- **Hot Streak.** ALERT A wallet with score ≥0.65 is on a 5-day winning streak with a risk-adjusted return proxy ≥0.6. Streak data is daily-resolution; intraday breaks do not update until the next scan.
- **Alpha Confirmation.** ENTRY Any other recipe's signal is corroborated by 3 or more wallets with score ≥0.75 holding fresh exposure in the same direction. The original signal is removed; double-counting is prevented.
- **Smart Exit.** EXIT A wallet with score ≥0.65 cuts ≥20% of its exposure while its regime alignment score falls below 0.35.
- **Patience Trap.** ALERT A wallet with score ≥0.65 holds an underwater position (>5%) for two or more consecutive cycles without reducing.

SIGNALS

SIMPLE

When several of our proven wallets all buy the same coin at the same time, that is unusual enough to notice. We call that a signal. It is a factual observation about what just happened, not a forecast of what will happen next.

TECHNICAL

13 detectors diff consecutive position snapshots. Whale Convergence fires when 3+ wallets add combined ≥$500K in the same direction within 5 minutes. Alpha Confirmation re-emits any detector's signal when corroborated by 3+ wallets with score ≥0.75, removing the original to prevent double-counting.

4.3 Confidence (EV) Scoring

Every per-wallet signal carries a confidence score between 0 and 1, derived from the signaling wallet's own 30-day trade history. The score combines win probability, average winning trade, average losing trade, and estimated execution cost (Hyperliquid taker fee plus live order-book slippage on a $10K reference notional). A 5x expected return relative to average loss maps to 1.0. Scores above 0.6 are considered actionable.

Slippage estimates are computed against the live order book for the ten most-traded coins in the cohort each cycle. Less liquid coins use the fee-only cost, which understates true execution cost.

Group-level signals carry no confidence score since there is no single wallet track record behind them. Newly activated wallets with no completed performance history carry no score either. Both states are labeled explicitly in the feed.

CONFIDENCE SCORE

SIMPLE

Based on how that trader performed over the last 30 days, we estimate whether following this signal would make money after fees. We show the result as a number from 0 to 1. Above 0.6 means we consider it worth acting on, based on that wallet's own history.

TECHNICAL

Each per-wallet signal carries a score in [0, 1] built from that wallet's own 30-day win rate, average win, average loss, and execution cost (0.035% taker fee plus order-book slippage on a $10K reference notional). A 5x expected return over average loss maps to 1.0. Above 0.6 is actionable. Group-level signals and wallets without a completed backtest carry no score.

SECTION 5

Quality Assurance

The feedback loop and the integrity layer

A signal detection system without outcome tracking is a guess with a name. Two processes close the loop.

Signal Validation

Every fired signal has its outcome measured against subsequent price action. The measurement window is configurable per recipe. A nightly learning job aggregates these outcomes per recipe and per wallet, producing a reliability stat that the ranking layer uses to reduce the weight of detectors or wallets whose historical signals have not paid out.

Without this feedback, a broken detector fires indefinitely. With it, underperforming detectors are muted automatically and high-performing wallets are surfaced more prominently.

Dashboard Truth

The second process enforces that the UI only promotes signals and wallets backed by sufficient validated history:

- **New wallets** (no completed daily scan) are tagged as unscored. Their signals appear in the feed without a confidence score.
- **New detectors** (insufficient signal history to compute reliability) are labeled accordingly.
- **Group-level signals** and signals without measurable outcomes are tagged as unvalidated.

This is an integrity layer, not a hiding layer. Raw data remains accessible on detail pages. The dashboard does not suppress data; it contextualizes it.

THE FEEDBACK LOOP

SIMPLE

After every signal fires, we check whether the price actually moved in the right direction. Over time, we automatically reduce the prominence of detectors that keep being wrong. The system corrects itself rather than relying on manual review.

TECHNICAL

Every fired signal's outcome is measured against subsequent price action. A nightly job aggregates those results per detector and per wallet, down-weighting ones whose historical signals have not paid out. The dashboard only promotes wallets and detectors backed by enough validated history; signals without measurable outcomes are labeled, not hidden.

SECTION 6

Infrastructure

GitHub Actions for batch work, Vercel for client-facing endpoints

DAILY SCAN - MIDNIGHT UTC, 50-MINUTE BUDGET

Pulls candidates from the public leaderboard and from recent trading activity on the most active coins. Runs each candidate through the nine quality gates, checks for manipulation patterns across the last 30 days of fills, builds trading profiles for all passing wallets, computes 30-day performance records from closing fills, and aggregates detector accuracy from historical signal data. Uploads a scan summary artifact with a 7-day retention window.

SIGNAL DETECTION - CONTINUOUS, 10-SECOND BUDGET PER CYCLE

A GitHub Actions workflow pings the detection endpoint continuously, keeping it warm. Each cycle loads a rotating window of active wallets, fetches their current positions and market data, computes regime and scores, runs all 13 detection recipes against consecutive position snapshots, calculates confidence scores using live order-book data for the ten most-traded coins, and writes a fresh cache payload. Stale wallet pruning and health checks run in the background after the main cycle completes.

SIGNAL LEARNING - 01:00 UTC DAILY

Measures the price outcome of every unresolved past signal, updates per-detector and per-wallet accuracy aggregates, and uploads a learning summary artifact with a 14-day retention window.

CLIENT - 60-SECOND POLL

The browser polls for updates every 60 seconds and reads the cached snapshot. It triggers a background refresh if the data is more than 5 minutes old. A separate monitoring workflow runs every 15 minutes and raises an alert if the cache has not been refreshed within 20 minutes.

SECTION 7

Known Limitations

An honest list. Any of these could mislead a user if not stated.

01 **Liquidation detection is approximate.** The Liquidation Flush detector uses cohort notional shrinkage combined with a price movement threshold as a proxy for liquidation. True real-time detection requires the Hyperliquid WebSocket liquidation feed, which is incompatible with the current serverless architecture. On the roadmap.

02 **Regime is BTC-only.** A wallet correctly positioned in a trending altcoin while BTC is ranging receives a below-average regime alignment score. There is no per-coin or sector regime. On the roadmap.

03 **Confidence scores use a fixed $10K reference notional.** The score does not scale to the signaling wallet's actual position size. A $5M account and a $50K account receive the same slippage estimate for the same coin. On the roadmap.

04 **Slippage is zero for coins outside the top 10.** Only the ten most-traded coins in the cohort have their order books walked each cycle. All other coins use fee-only cost, which understates true execution cost on less liquid names.

05 **Scoring weights are assumed, not calibrated.** The 35/25/25/15 factor split is a reasonable starting point that has not been empirically fit to forward performance. A calibration script exists; the optimization is not yet complete.

06 **New wallets have no confidence score.** Until the nightly scan produces a completed performance record, signals from newly activated wallets fire without a score. Marked explicitly in the feed.

07 **Hot Streak data is daily.** Intraday streak breaks do not update until the next scan. The UI shows a staleness indicator to communicate this.

08 **Leaderboard discovery fallback is partial.** If the Hyperliquid leaderboard API fails, discovery falls back to fills mining only, covering the top-10 coins but missing wallets active in less liquid markets.

WHAT THE SYSTEM DOES NOT DO

SIMPLE

It does not predict prices. It does not tell you what to trade. It tells you what a group of proven traders just did. What that means for your specific situation is a question the system cannot answer, and it will not pretend otherwise.

TECHNICAL

No directional prediction. No position sizing advice. Confidence scores are historical estimates, not forward guarantees. Slippage is understated for illiquid coins. Liquidation detection is approximate (the serverless architecture cannot tap the real-time liquidation feed). Newly activated wallets fire without a score. Regime classification is BTC-only. Scoring weights are unoptimized.

SECTION 8

Roadmap

Items ordered roughly by anticipated delivery sequence

SPRINT 5 - ACTIVE

Wallet Alerts and Paper Copy Trading

Per-wallet signal subscriptions with configurable delivery. Paper copy trading lets users track hypothetical positions copied from high-score wallets, with P&L tracked in isolation from real portfolios.

SPRINT 6 - PLANNED

Authenticated User Accounts

User accounts, alert persistence, and portfolio tracking. Authentication enables personalization and notification delivery across sessions.

ARCHITECTURAL IMPROVEMENTS - PARALLEL TRACK

- **Per-coin regime.** Replace the BTC-only assumption with a per-asset or sector-level classifier.
- **WebSocket liquidation feed.** Replace the approximate detector with direct real-time liquidation data from Hyperliquid.
- **Position-scaled confidence.** Replace the fixed $10K reference notional with the signaling wallet's actual position size.
- **Scoring weight calibration.** Fit the 35/25/25/15 split to empirical forward performance data.

SPRINT 8 - BLOCKED (DATA GATE: 100+ MEASURED SIGNAL OUTCOMES, 30D HISTORY)

Autonomous Documentation Agent

An agent that watches the codebase for behavioral drift against this whitepaper and the product overview. Phase 1: drift detection, opening a GitHub PR when a gate threshold, recipe rule, or data-flow step changes. Phase 2: auto-patch documentation based on code changes. Blocked pending sufficient historical signal outcome data.

SECTION 9

Glossary

Terms as used in this document

Active cohort

> Wallets currently passing all activation gates and all continuous hygiene checks. Typically ~500.

Closing fill

> A fill that closes or reduces a position at a realized profit or loss. The only source of realized PnL. Unrealized PnL is never counted.

Daily PnL array

> A 30-element vector of per-day realized PnL, index 0 = 30 days ago. Source of the risk-adjusted return, PnL consistency, and drawdown scores.

EV score

> Expected value normalized to [0, 1], where a 5x expected return over average loss equals 1.0. Considered actionable above 0.6.

Hygiene gate

> A per-cycle health check (quiet wallet, equity collapse, no recent fills) that deactivates wallets between daily scans without waiting for the next batch run.

Regime

> BULL, BEAR, or RANGING. Derived from the BTC 24-hour return crossing plus or minus 1%. No per-coin regime exists currently.

Regime fit

> Alignment between a wallet's directional bias (net long vs. net short) and the current BTC regime. A flat wallet scores 0.5.

Signal event

> An output of one of the 13 detection recipes for a given cycle. Persisted with its confidence score and validation state.

Signal outcome

> The measured price result of a fired signal, recorded after the fact and fed into the nightly learning job.

Snapshot pair

> Two consecutive position records for one wallet from back-to-back detection cycles. All 13 signal recipes diff them to detect position changes.

Wash / sybil cluster

> A group of wallets flagged by the manipulation detector as trading with themselves or each other in a coordinated pattern that would distort signal detection.

Waterproofing

> Collective term for the three quality layers that run after activation: behavior profiling, signal outcome validation, and truthful surface gating. Prevents unvalidated signals from being presented as settled fact.